

DESIGN THINKING LAB REPORT- 2

Topic:

Electricity outage information is not communicated clearly.

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STEP 1: Define the Target User

Target Users:

- Age: 18–60 years
- Education: School / College / Working professionals
- Device: Smartphone
- Digital Literacy: Medium
- Services Used: Electricity board app, SMS alerts, WhatsApp groups, websites

Problem Context:

Users do not receive **clear, timely, and understandable information** about power cuts.

STEP 2: Initial Assumptions

ID

Assumption

A1 Users do not get outage alerts on time

A2 Messages are confusing or incomplete

A3 Users check multiple sources for confirmation

A4 Apps/websites are difficult to navigate

A5 Users want exact restoration time

STEP 3: Interview Questions

1. How do you usually know about a power cut?

Mostly through neighbors, sudden power loss, or checking social media/WhatsApp groups.

2. Do you receive any SMS or app notification?

Rarely. Notifications are either delayed or not received at all.

3. **What problems do you face during electricity outages?**
Work disruption, device charging issues, internet loss, and difficulty planning daily activities.
4. **Is the restoration time clearly informed?**
No. Restoration time is usually unclear or keeps changing.
5. **Which method do you prefer for outage updates?**
Real-time SMS or mobile app notifications from the official electricity provider.
6. **What improvement would you like in outage communication?**
Advance alerts, accurate restoration time, and one reliable official communication platform.

STEP 4: User Interview (Sample)

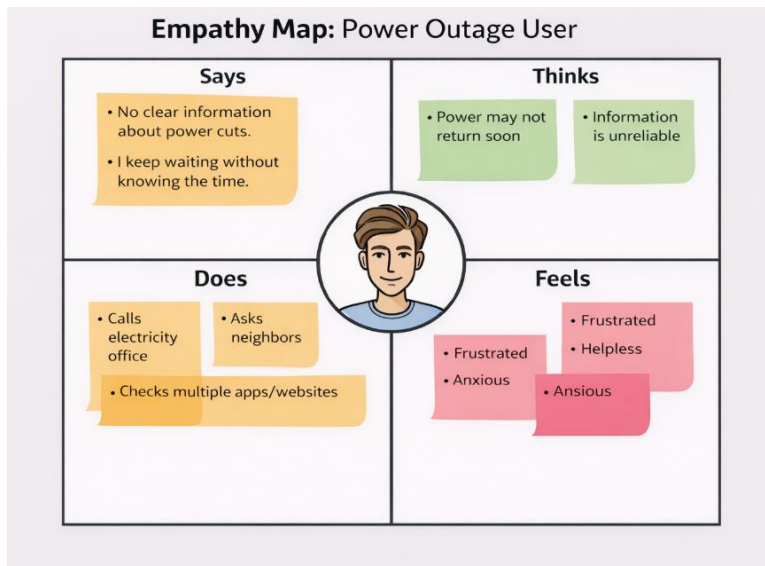
Observations:

- User repeatedly checks phone for updates
- Shows frustration due to uncertainty
- Searches Google or asks neighbors

User Statements (Quotes):

- “I never know how long the power cut will last.”
- “SMS comes late or not at all.”
- “Website is confusing to check.”

STEP 5: Empathy Map



STEP 6: Validate Assumptions

Assumption	Validated?	Reason
Alerts not received on time	✓	User reported delay
Messages are confusing	✓	User couldn't understand info
Checks multiple sources	✓	Observed behavior
App difficult to use	⚠️	Mixed response
Needs exact restoration time	✓	Strong user need

STEP 7: User Insights

- Users need **clear and timely outage communication**.

- **Uncertainty of restoration time** causes frustration.
- **Trustworthy single source of information** is important.

✗ Users don't care about outage details

✓ Users want **accurate time and clear updates**

STEP 8: User Persona



Priya Sharma
IT Professional (Works from Home)

“Unplanned power outages throw off my entire schedule. I need clear, early warnings so I can manage my work and family without stress.”

Priya Sharma
IT Professional (Works from Home)

Bio: 32

Location: Urban residential area

Tech Level: Proficient in using apps and online tools

Goals

- Plan work and daily activities during outages
- Receive clear, advance notifications about power cuts

Frustrations

- Frequent sudden power cuts with no warning
- Conflicting or vague information from different sources

Goals

- Timely SMS/app alerts before outages
- Accurate start and end time information

Needs

- Timely SMS/app alerts before outages
- One-stop official outage communication source



Arjun Mehta
Financial Analyst (Works from Office)

“ I need reliable alerts before important market events to make informed decisions and stay ahead of potential risks.

Arjun Mehta
Financial Analyst (Works from Office)

Bio: 33

Location: Downtown business district

Tech Level: Advanced, proficient in using financial software and spreadsheets

Goals - Receive timely alerts before market events - Get accurate, real-time updates on market conditions	Frustrations - Unexpected market fluctuations without prior alerts - Outdated or delayed market information
Goals - Timely SMS/app notifications for market events - Reliable and up-to-date real-time market information	Needs - Timely SMS/app notifications for market events - Reliable and up-to-date real-time market information

STEP 9: Translate Insights to Software Needs

Insight

Need clear alerts
Wants restoration time
Confusion in website
Needs single reliable source
Frequent checking

Software Requirement

Instant push notification & SMS
Show exact estimated time
Simple, clean interface
Centralized outage dashboard
Live status updates

STEP 10: Reflection

1. What surprised you?

Users feel more stress from **uncertainty** than the outage itself.

2. Which assumption was wrong?

Not all users find apps difficult; the issue is mainly **unclear information**.

3. How did empathy change thinking?

Focus shifted from **technical outage** to **user communication clarity**.

4. Effect on software design?

Design must prioritize:

- Clear language
- Real-time updates
- Accurate restoration time